

Project Plan

Document Number: [01]

Friday, September 25th, 2026

KSU Legacy Brick Locator

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This form must be completed in accordance with the Software Engineering Body of Knowledge (SWEBOK).

Table of Contents

1. Introduction	3
2. Project Overview	3
2.1 Scope	3
2.1.1 Identification.....	3
2.1.2 System Overview.....	3
2.1.3 Document Overview.....	3
2.2 Current System or Situation	3
2.3 Background, Objectives, and Scope.....	3
2.4 Operational Policies and Constraints	3
2.5 Description of Current System or Situation	3
2.6 Users or Involved Personnel	4
2.7 Standards to follow.....	4
3. Development Background/Approach	4
3.1 High Level Estimates	4
3.2 Key Contacts and Stakeholders.....	4
4. Features, Primary Deliverables, and External Commitments.....	4
4.1 Feature List.....	4
4.2 Customer Deliverables	4
5. Project Schedule	5
5.1 Major Project Milestones	5
5.2 Project Status Tracking & Working Meeting Minutes.....	5
6. Project Work and Product Estimates	5
6.1 Estimate Summary	5
7. Project Resource Requirements.....	6
7.1 Staffing/ Skill Requirements	6
7.2 Plan to Fill Skill Gaps	6
8. Dependencies and Constraints	6
8.1 Constraints.....	6
9. Risk Management.....	7
9.1 Risk Management Strategy	7
9.2 Initial Risk List.....	7
10. Project Configuration and Data Management	7
10.1 Configuration Management.....	7
11. Project Process.....	7
11.1 Software Life Cycle Model	7
12. Referenced Documents	7
13. Glossary.....	7
14. Change Record	7

1. INTRODUCTION

This Project Plan describes the scope, context, stakeholders, activities, resources, schedule, constraints, risks, quality goals, and processes. It is intended to provide a guide to how development work will be managed.

The scope of this project covers a few necessary features. The project shall be able to grab information about a given brick from the database. A user can look at the information and inscriptions on a brick. Directions to a given brick shall be provided in the form of maps of a given section.

2. PROJECT OVERVIEW

2.1 Scope

2.1.1 Identification

This document applies to the KSU Legacy Brick Locator (LBL), identified as P16-02, Version 1.0 (v1.0), Release 1.0, a capstone project developed for Kennesaw State University's Software Engineering Program and the Division of University Advancement.

2.1.2 System Overview

The LBL is a web-based application designed to provide KSU Legacy Brick purchasers with an effective means of locating their bricks on either of KSU's campuses. The LBL originated as a senior capstone project at Kennesaw State University, with development beginning in the Fall 2026 semester. Kennesaw State University is both the sponsor and acquirer of the LBL and intended for use by both the KSU University Advancement division, and those prior faculty, staff, or alumni that have purchased a Legacy Brick from the institution. The LBL is developed by Aaron Smith, Cole Williams, Darion Smith, Jasper Zheng, and Olivia Tyson, guided by Professor Yan Huang. The project will be hosted locally throughout development, with deployment to AWS (Amazon Web Services) by project handoff. Support has yet to be determined and will depend on decisions made by KSU University Advancement.

2.1.3 Document Overview

This document summarizes the scope, milestones, methodology, resources allocation, and risk management strategies for the KSU Legacy Brick Locator (LBL). It will serve as the blueprint for development and guide the team throughout the three-milestone goal.

2.2 Current System or Situation

Currently, no automated system exists, and the client is required to manually access and find the location of bricks via a manually assembled and maintained spreadsheet and then must communicate to the user the location of the legacy brick. This is a relatively time-consuming task and could easily become infeasible with the purchase of more bricks.

2.3 Background, Objectives, and Scope

Background: KSU offers supporters the opportunity to purchase a commemorative brick that is placed within two locations on both the Marietta and the Kennesaw Campus.

Objectives: The goal is to allow users to search for their brick through a web application and allow the administrators to input brick data. The platform will reduce staff workload and foster donor engagement.

Scope: Covers public search, interactive location mapping, brick detail views, secure administrative CRUD operations, CSV/Excel data migration tooling, and administrative reporting. Excludes any sort of payment processing (the user will be sent to the official site) and native mobile apps.

2.4 Operational Policies and Constraints

Operational Policy: All application codes and documentation will be maintained in a private GitHub repository shared with the students. Administrative changes require confirmation or soft deletions to preserve data integrity. Constraints: The project must meet all the requirements listed by the stakeholders and be completed within a single academic semester. Development is restricted to the agreed upon technology stack.

2.5 Description of Current System or Situation

There is currently no automated system for storing and retrieving the location of KSU Legacy Bricks. The current situation is entirely manual, relying on students to compile a spreadsheet with the brick names, inscriptions, and location information. When a user/purchaser of a brick would like to locate a brick, KSU University Advancement staff must manually open and search for the brick record and then communicate that information to the user requesting the location.

The operational environment includes both sides where bricks are installed on each of KSU's campuses. Major components of the existing system include the bricks installed on either of the two KSU campuses, the spreadsheet in which the records for installed bricks are compiled, KSU University Advancement staff, and prior KSU alumni, faculty or staff who have purchased a legacy brick.

The current system risks struggling with speed and throughput as total volume of legacy bricks sold increases, due to a relatively time-consuming and error-vulnerable process for retrieving the location information.

The current system risks shortcomings regarding several quality attributes, as described below:

- Maintainability: Due to the manual nature of the current system cataloging process, maintenance of the spreadsheet may become difficult to maintain as more bricks are sold, moved, or removed.
- Availability: Since the system relies heavily on KSU University Advancement for communication and retrieval, availability is restricted to business hours of KSU staff and further limited by their capacity to respond amongst other job requirements.
- Efficiency: The current system struggles with turnaround time due to the manual lookup process, which must compete with staff's other job responsibilities along with lookups simply being a relatively time-consuming process.

2.6 Users or Involved Personnel

Guest Visitors (Alumni, Donors, Families, Campus Visitors): Unauthenticated users looking to locate bricks easily via desktop or mobile devices. They require intuitive search tools, clear inscriptions, and simple accessibility.

Administrators (KSU Staff): Authorized personnel who log into a secure dashboard to manage brick records, import spreadsheets, handle duplicates, and export reports.

Development Team (Aaron Smith, Cole Williams, Darion Smith, Jasper Zheng, Olivia Tyson): Undergraduate students responsible for full-stack engineering, documentation, testing, and delivery.

2.7 Standards to follow

Standard: ISO/IEC/IEEE 12207 (Systems and software engineering — Software life cycle processes).

Importance: This standard is critical because it provides an internationally recognized framework for managing software processes, requirements definition, design, testing, and maintenance. Adhering to it ensures that the LBL project follows rigorous software engineering principles appropriate for an institutional capstone deliverable.

3. DEVELOPMENT BACKGROUND/APPROACH

The LBL will be developed as a web application intended to be accessible to both mobile and desktop users. The system will be developed under a three-tiered architecture including:

A React-based user interface utilizing Mapbox and the Mantine UI Toolkit.

A Java Spring REST API backend, utilizing the components: Spring MVC, Spring Data JPA, Spring Security, and developed with Lombok and Maven.

A PostgreSQL server and database contained in Docker throughout development.

The development team consists of 5 students working under the KSU Division of University Advancement, the sponsor of the project. Members are divided by technical role between Frontend and Backend. The majority of team members have general familiarity with front-end development, React, and Java.

3.1 High Level Estimates

Total Effort: ~300–400 person-hours across the team for one semester.

Lines of Code (LOC): Estimated 4,000–7,000 LOC (Full-stack React frontend and backend API).

Documentation: ~40–60 pages across milestones (Project Plan, SRS, SDS, Test Document, User/Admin Guides).

Test Cases: ~40–60 functional and integration test cases covering search matching, data import validation, and role-based permissions.

Duration: 15 weeks (One Semester: Fall 2026)

3.2 Key Contacts and Stakeholders

Sponsor Organization: Kennesaw State University – Division of University Advancement

Primary Sponsor Contact: Madison Snyder (Parent and Family Engagement Manager) | Email/Phone: mrishop@kennesaw.edu

Faculty Advisor: Professor Yan Huang | Email/Phone: yhuang24@kennesaw.edu

Development Team:

Aaron Smith (Developer / Team Member)

Cole Williams (Developer / Team Member)

Darion Smith (Developer / Team Member)

Jasper Zheng (Developer / Team Member)

Olivia Tyson (Developer / Team Member)

4. FEATURES, PRIMARY DELIVERABLES, AND EXTERNAL COMMITMENTS

4.1 Feature List

- Brick Lookup: Users will be able to search for their purchased brick by name and description, as well as filter results by section, row, and campus.
- Brick Preview: Users will be able to preview a digital representation of their brick similar to its actual appearance on KSU campus grounds.
- Brick Location Preview: Users will be able to review the location of their brick via KSU provided diagrams along with geographical map representations, provided through the implementation of Map Box
- Directions: Users will be able to request directions to their specifically selected brick and export those directions to their preferred mapping application (Google Maps, Apple Maps).
- Administrative Platform: The system will include a separate administrative platform by which to manage relevant data.
- Brick Cataloging: Properly authorized users will be able to manually catalog installed bricks and add them directly to the database for lookup via the Administrative Platform.
- Brick Editing: Properly authorized users will be able to edit and remove brick records as necessary to correct misspellings, incorrect information, or remove bricks if needed.
- Spreadsheet Import: The system will, via the administrative platform, allow for properly authorized users to upload a spreadsheet, which will then be ingested by the system and appended to the database's records.

4.2 Customer Deliverables

Date	2026-09-28	2026-10-26	2026-11-30
Deliverables	Project Plan Document Formal Requirements Document Stakeholder Interview Summary Source-Spreadsheet Data Assessment Public vs. Restricted Field Classification Proposed Database Schema and ERD Search Requirements Brick-Location representation strategy User Stories and Acceptance Criteria Wireframes and Interactive Mockups Architecture Diagram Private GitHub repository	Database implemented Sample brick data imported Public search interface Search-results and brick-detail views Initial location/map experience Secure administrator login Administrative create, update, and deactivate functions Sponsor demonstration Updated requirements and technical documentation	Final interactive location experience Excel/CSV import validation Administrative reports Advancement engagement features Responsive and accessible interface Security and privacy review Test plan and test results Setup and maintenance documentation User and administrator guide Final presentation Complete GitHub handoff Future-development recommendations

5. PROJECT SCHEDULE**5.1 Major Project Milestones**

Date (YYYY-MM-DD)	Milestone/ task	Deliverable	Remarks
2026-09-28	Milestone 1	Initial Setup and Documentation	Project Plan, Repository Prepared, Roles Assigned
2026-10-26	Milestone 2	Foundational frontend interfaces REST API Complete	Necessary frontend elements implemented and connected to API endpoints
2026-11-30	Milestone 3	Full application	Should be ready for deployment, if not deployed to AWS.

5.2 Project Status Tracking & Working Meeting Minutes

Team meetings will be hosted weekly, with attendance mandatory for all team members.

Topics will include progress on current deliverables with active review, discussion about future deadlines and deliverables, and discussion about current roadblocks, risks, and schedule impacts.

6. PROJECT WORK AND PRODUCT ESTIMATES

Inputs to estimation include:

- 1) Deliverables listed in [Section 4](#)
- 2) Major Milestones listed in [Section 5](#)
- 3) Tasks and/or sub-tasks identified in the detailed project Schedule.

6.1 Estimate Summary

Estimates are based on deliverables and milestones from sections 4 and 5, respectively.

The tables below summarize the product size and effort estimates:

Project	Estimate Attributes		
	Size		
WBS areas	Unit of Size	Size	Effort
Total Requirements Effort (includes feature-related <u>and</u> "other" (non-feature) Requirements work)			40
Feature Related Requirements Size and Effort Totals	PAGES	15	20
Total Development Effort (includes feature-related <u>and</u> "other" (non-feature) Development work)			200
Feature Related Development Coding Size and Effort Totals	LOC	7,000	180
Feature Related Development Documentation Size and Effort	PAGES	5	20
Total Testing Effort (includes feature-related <u>and</u> "other" (non-feature) Testing work)			40
Feature Related Testing Size and Effort Totals	TEST CASES	30	30
Feature Level Effort Total (from Feature Estimate Worksheet)			250
Development Effort Total (Includes Feature Level and project level overhead for Requirements, Development, and Testing)			280
Project Level Effort Total (from Project Level Effort Estimates worksheet, excluding requirements, development, and testing)			35
Project Total Effort (Project Totals + Feature Totals)			315

7. PROJECT RESOURCE REQUIREMENTS

7.1 Staffing/ Skill Requirements

Role: Team Leader – Aaron Smith

Critical Skills: React, JavaScript, Java, Spring Boot, PostgreSQL, Docker, AWS

Skill Gaps: None Significant

Role: Development Manager – Darion Smith

Critical Skills: React, JavaScript, Java

Skill Gaps: PostgreSQL, Docker, AWS

Role: Planning Manager – Cole Williams

Critical Skills: React, JavaScript, Java, AWS, Planning

Skill Gaps: PostgreSQL, Mapbox, Spring Boot

Role: Quality & Process Manager – Olivia Tyson

Critical Skills: Software testing, Quality Assurance, JUnit, Java, Git/GitHub, Requirements Analysis, Documentation

Skill Gaps: Spring Boot, PostgreSQL, Docker, AWS

Role: Support Manager – Jasper Zheng

Critical Skills: Java

Skill Gaps: PostgreSQL, Docker, JavaScript, React, JavaScript, AWS

7.2 Plan to Fill Skill Gaps

Cole Williams – Read documentation on PostgreSQL and Spring Boot. Become more familiar with MapBox maps and geo located zones.

Darion Smith – Familiarize myself with documentation of Both PostgreSQL as well as Spring Boot. Review information regarding all relevant tools for this project.

Jasper Zheng – Bridge frontend gaps by learning JavaScript and React basics. Learn PostgreSQL for basic knowledge of database management. Explore Docker fundamentals.

Olivia Tyson – Learn enough about each technology to test, troubleshoot, and verify work that has been completed. Work on the side of team members to gain practical experience with Spring Boot, PostgreSQL, Docker, and AWS. Apply these concepts during project testing and development.

8. DEPENDENCIES AND CONSTRAINTS

8.1 Constraints

- The project must be completed by the end of the Fall 2026 semester
- Development is restricted to the previously determined technology stack
- Team availability is limited by coursework, employment, and/or general scheduling conflicts

9. RISK MANAGEMENT

9.1 Risk Management Strategy

9.1.1: The team will verify schedules ahead of time and adjust meetings and communication accordingly.

9.1.2: The team will adhere strictly to the requirements as defined in the project specification and SRS, staging all expansion of features beyond requirement definitions in branches, contingent on the acceptance of the sponsor.

9.1.3: The team will adhere to a strict branching policy when working on and submitting changes to the repository, branching first by front-end/back-end/database, and further branching by feature.

9.2 Initial Risk List

This is the initial risk list; the risks are listed in priority order from top to bottom. Descriptions are provided below the table. Risks with a pre-mitigation magnitude of 2.0 or below are not listed.

Risk number	Risk Priority (H, M, L)	Likelihood of Occurrence	Risk name: brief description	Mitigation Strategy “ACCEPTED” or “MITIGATED” with pointer to plan.
1	M	Medium	Schedule Conflicts and Changes	Mitigated (See 9.1.1)
2	H	Medium	Workload Management	Accepted
3	L	Low	Scope Creep	Mitigated (See 9.1.2)
4	M	Low	Submission Conflicts	Mitigated (See 9.1.3)

10. PROJECT CONFIGURATION AND DATA MANAGEMENT

10.1 Configuration Management

The project will be stored on a GitHub repository and will follow strict branching procedures (see 9.1.3). Submissions will only be accepted after review. Environment configuration will be managed through Docker throughout development, with deployment configuration for AWS planned at project handoff.

11. PROJECT PROCESS

11.1 Software Life Cycle Model

The team will adopt an Agile methodology, adapted to match pace with the four-milestone structure laid out by the project specification.

12. REFERENCED DOCUMENTS

Software Requirements Specification [02], revision 1, 2026-09-22
 Software Design Specification [03], revision 1, date TBD
 Project Specification [00], revision 1, 2026-09-01

13. GLOSSARY

LBL: Acronym referring to the KSU Legacy Brick Locator
 SRS: Acronym referencing the Software Requirements Specification
 SDS: Acronym referencing the Software Design Specification

14. CHANGE RECORD

2026-09-25 – Initial Document Creation – Aaron Smith